Biomedical Image Analysis Project SS2026

Kickoff Meeting

Mental model and BIMAP aims

You will learn to think like a researcher, not how to become a prompt engineer!

Your project should look like a small NeurIPS paper, not a Kaggle notebook.

Projects are designed being fun and ambitious and they need your time (10 ECTS ~ 300 working hours)

BIMAP Success Stories

- Many students continued project as MSc thesis in the lab
- People wrote or write papers about their BIMAP project
  - Resource-efficient training and inference of TransUNets (BIMAP 2025, in preparation)
  - Neuro-SAM - segmentation foundation models for neuroscience (BIMAP 2024, in prep)
  - Rodríguez, B. L., Borges, G. L., García, M. L. B., Schützenberger, A., & Kist, A. M. (2024, May). Neural Radiance Fields for 3D Reconstruction in Monoscopic Laryngeal Endoscopy. In 2024 IEEE International Symposium on Biomedical Imaging (ISBI) - BIMAP 2023

Seminar overview

Phase 0

Hustle Sprint
(next two weeks)

Goal: MVP

Gate to stay in project

Phase 1

Formalizing

Formulate Hypotheses
Fix evaluation protocol
Define experiments

Phase 2

Controlled Experiments

Baselines, Ablations,
HPO, keep protocol log

HACKDAY I

Phase 3

Consolidation

Testing reproducibility
Unify plots/final figures

HACKDAY II

Phase 4

Finish line

Hand in report, code
and give final
presentation

Timetable

Meetings BIMAP General Assembly

14.04.2026	Tuesday	16.00-17.30	Kickoff	Explaining course and pitch projects
Phase 0: Hacking time, preparing MVP and presentation				
24.04.2026	Friday	12.15-13.45	MVP Presentations	Phase 0 achievers stay
Phase 1: Formulate Hypotheses, fix evaluation protocol, define experiments				
08.05.2026	Friday	12.15-13.45	Phase 1 - Interim presentation	
Phase 1 Reiteration				
22.05.2026	Friday	12.15-13.45	Phase 1 - Interim presentation, updates	
Phase 2: Controlled Experiments				
29.05.2026	Friday	12.15-13.45	Phase 2 - Interim report	
Phase 2: Controlled Experiments				
12.06.2026	Friday	14.00 - 18.30	Hackday I (Dinner at 18.30)	
Phase 2: Controlled Experiments				
26.06.2026	Friday	12.15-13.45	Phase 2 - Interim report	
Phase 3: Consolidation				
03.07.2026	Friday	14.00 - 18.30	Hackday II (Dinner at 18.30)	
Phase 4: Finalizing				
14.07.2026	Tuesday	16.00-17.30	Phase 4 - Interim report, wrapping up	

Presentations: 23.07.2026	09.00-13.00		
Report due: 31.07.2026			

Alternative: July 2nd
June 29th

Science is key.

We enforce scientific methods.

This is not how it works:
"We tried many things."

"After tuning…"

"We experimented with different architectures."

"We achieved up to…"

Rather
"We evaluated the following controlled variants."

"We tested X while holding Y constant."

Every experiment must have:

1. Hypothesis
2. Single controlled change
3. Documented configuration
4. Run ID
5. Metric
6. Interpretation

Deliverables

- Attendance is mandatory
  (you can miss once, twice with a doctor’s note) → otherwise out
- Intermediate presentations
  (not graded, but have to be given, if sick you need alternative dates, we make your life easy and provide templates to be filled by you)
- Get things done (hacking day I)
- Reproducing code from other projects (hacking day II)
- For grade:
  - Prototype/code
  - Presentation (15 min)
  - Report (15 pages)

Prototype/Code

Needed for reproduction

- Version control (Github) is mandatory
- Experiment log is mandatory
- Documentation and comments are mandatory

- Unit tests depending on project
- Tutorial for newcomers depending on project
- Python package pip installable depending on project

Report

- 15 pages in JMLR template
- Report has to be written using LaTeX (Overleaf project)
- Content like a scientific paper
  - Abstract
  - Introduction
  - Methods
  - Results
  - Discussion
  - References

Experiments

Experimental log is mandatory.

Tier A (mandatory)

- Experiments.csv
- Config.yaml per run
- 3 seeds minimum per experiments

Tier B (optional)

- Use tools like wandb, MLflow, etc.

UID

csv

```
001,2025-04-12,CNN_small,MNIST,standard,42,lr=0.001;wd=1e-4,0.98,0.97,0.96,baseline
002,2025-04-13,CNN_small,MNIST,rotated,42,lr=0.001;wd=1e-4,0.98,0.60,0.58,distribution shift
```

Code

```
/experiments/
    config_001.yaml
    config_002.yaml
```

YAML

```yaml
model: CNN_small
learning_rate: 0.001
weight_decay: 1e-4
batch_size: 128
epochs: 20
seed: 42
```

Main deliverables and grading

Code

Reproducibility (5P)
Structure and Documentation (5P)
Evaluation correct (5P)
Experimental log (5P)

10P = 4.0

Report

Problem clear (5P)
Methods clear (5P)
Evaluation correct (5P)
Critical discussion (5P)

10P = 4.0

Presentation

Clarity (5P)
Technical depth (5P)
Reflexion (5P)
Q&A (5P)

10P = 4.0

Final grade = (Code + Report + Presentation) / 3

Hacking days

We will have two afternoons where we start at **2 PM** and hack until dinner at **6.30 PM**. We will provide dinner (pizza or similar, vegetarian or vegan possible).

The hacking days will have a clear structure.

- Aim for today (will be written on board)
- Hacking
- Results presentation (very short)
- Dinner
- More hacking or depart

Vibe coding

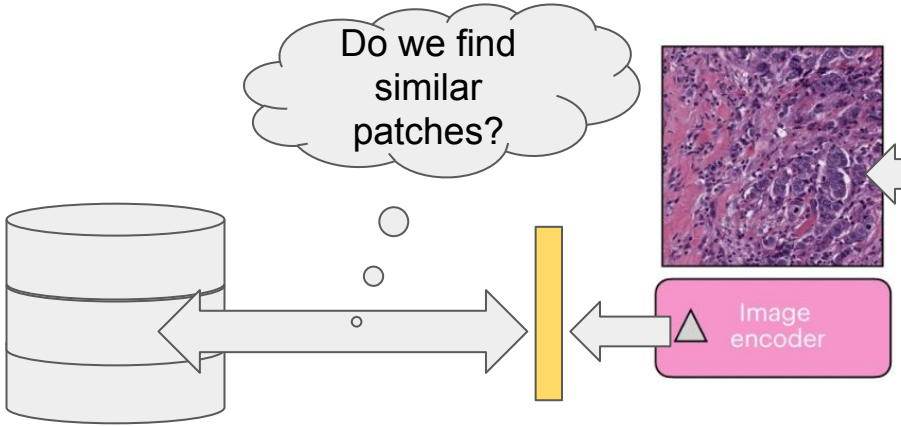
- It is not forbidden, but explicitly discouraged
- You need to follow the scientific principle and not vague descriptions
- The team will do code reviews with you - you need to be able to explain each single line of code *(high level, e.g. what does np.dot exactly here - e.g. computing the local attention mask, not how np.dot works under the hood)*

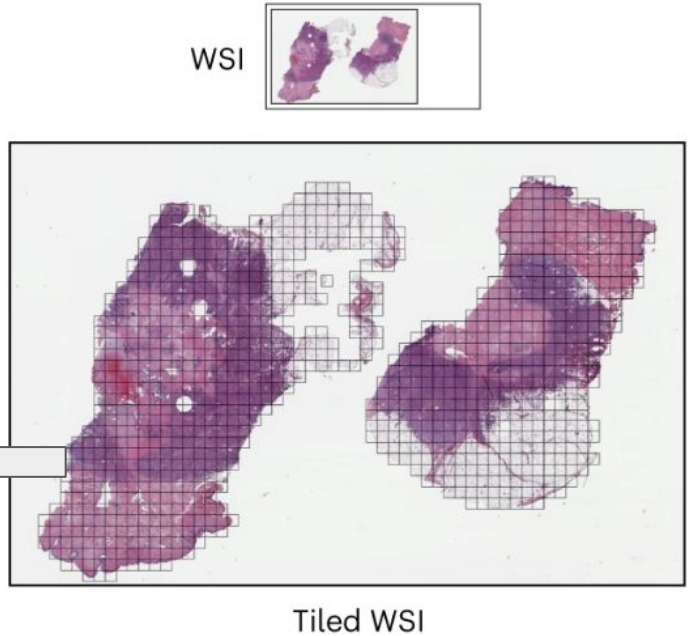
Vibe	Science
"We improved it."	"Run 14–16 show +3.2 ± 0.4% improvement."
"We tuned hyperparameters."	"We tested lr ∈ {1e-3,1e-4} with seed {1,2,3}."
"It works better."	"Statistically significant at $p < 0.05$."

Projects

HISTORAG

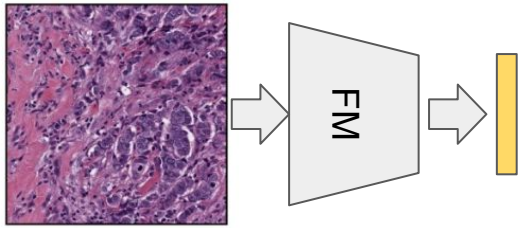
TL;DR: This project investigates whether retrieval-augmented methods can be used to build an interactive histopathology atlas from whole slide images (WSIs). Image patches extracted from slides are embedded using pretrained vision or vision-language models and stored in a vector database. The system should allow image or text queries to retrieve visually similar tissue regions and evaluate retrieval performance on labeled histopathology data.





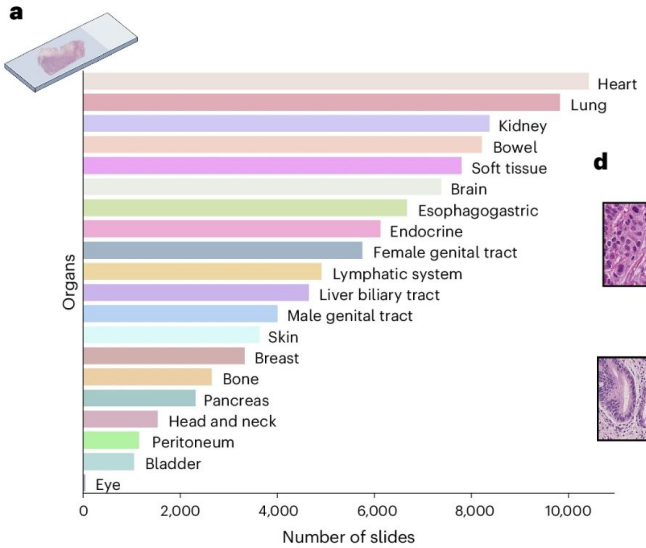
Lu et al., Nat Med 2024

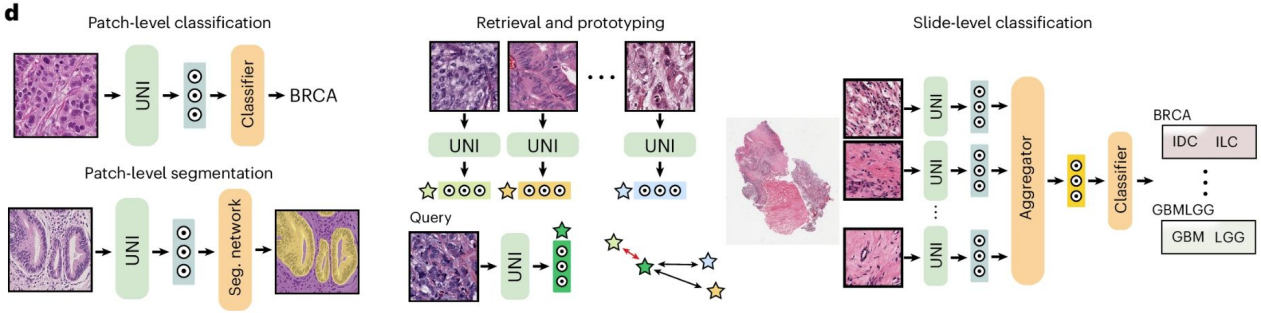
Foundation models



embedding

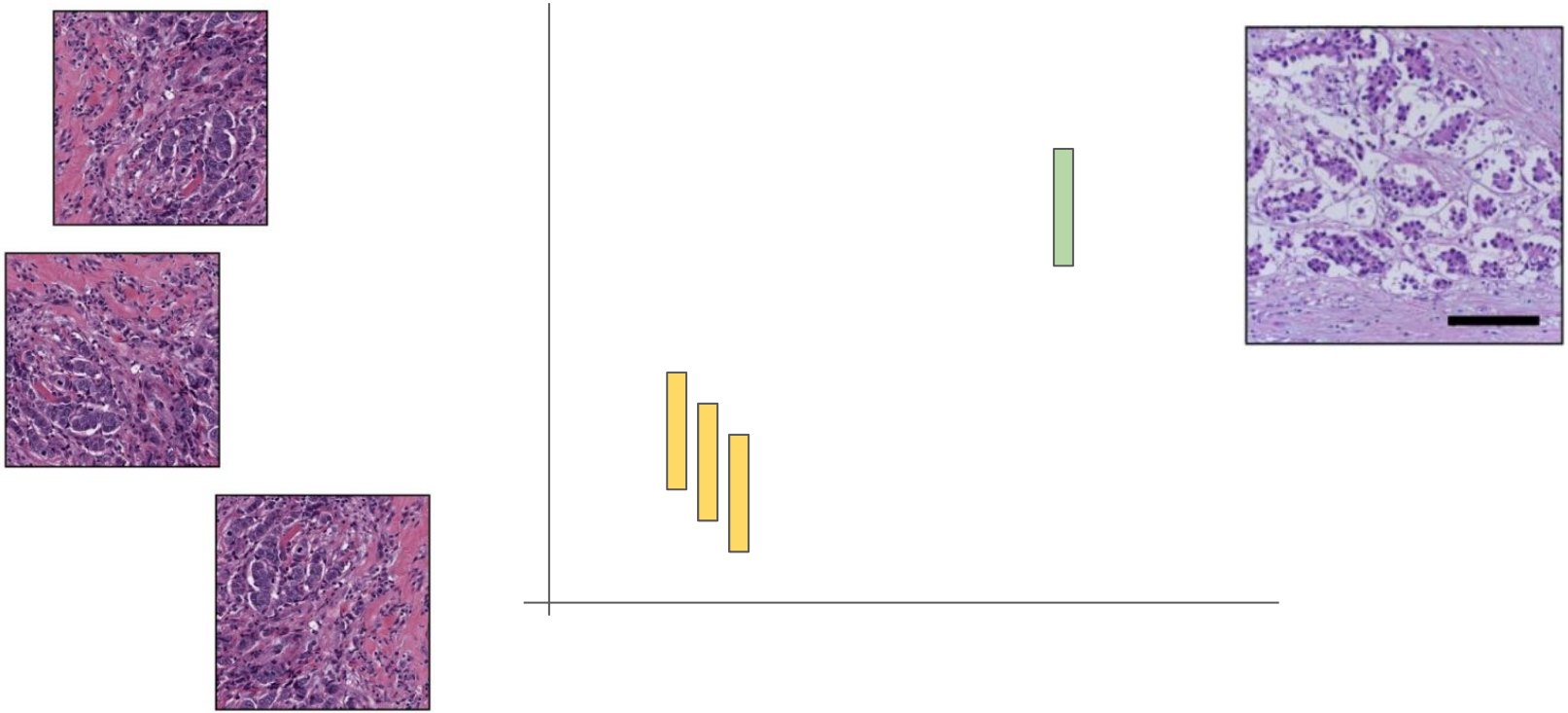
Condensation of 512x512 dimensions into 768 dimensions (~0.3%)





Chen et al., Nat Med 2024

We can also build an atlas



Which embedding is the “best”?

CLIP

TITAN

BEPH

UNI

ResNet

Phikon-v2

Virchow2

Atlas2

CONCH

Goal and MVP

Overall goal
Develop and evaluate an LLM-mediated, retrieval-based system for histopathology patches extracted from WSIs.

Minimal viable product (Phase 0 deliverable):
The MVP should implement a complete end-to-end pipeline that extracts image patches from whole slide images (WSIs), computes feature embeddings for each patch using a pretrained model, and stores these embeddings in a vector index for efficient similarity search. Given a query image patch, the system should retrieve the most similar patches from the database.

Assigned to **HISTORAG**:

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